Arithmetic and complexity

The least positive permanent of a sign matrix

Difficulty: challenging **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: Challenging because divisibility alone gives no construction attaining the bound for every order; specialist importance concerns exact permanent values and discrete matrix constructions.

Problem statement

For $n \geq 1$, let $\Omega_n = \{-1, 1\}^{n\times n}$ and define

$$\operatorname{per} A = \sum_{\sigma \in S_n} \prod_{i=1}^{n} a_{i,\sigma(i)}, \qquad p_n = \min\{\operatorname{per} A : A \in \Omega_n,\ \operatorname{per} A > 0\}.$$

The set in this minimum is nonempty, since the all-ones matrix has permanent $n!$. Is Kräuter's proposed formula

$$p_n = 2^{n-\lfloor \log_2(n+1) \rfloor}$$

valid for every n? The problem asks for attainability of the divisibility lower bound, with no rank restriction on A.

Relevance and ratings

This is an extremal question about a basic multilinear matrix function on discrete matrices. Exact values and divisibility constraints matter for permanent computation and constructions of sign matrices. The attainability problem across all orders merits challenging difficulty; its main audience is combinatorial matrix theory.

References

- I. M. Wanless, *Permanents of matrices of signed ones*, Linear and Multilinear Algebra **53** (2005), 427–433, §3, for attaining constructions through order twenty.
- D. Ingram and A. Razborov, *On the range of the permanent of* (± 1)*-matrices*, Linear Algebra Appl. 743 (2026), 271–285. The precise question is §6, Problem 3 in arXiv:2507.09433v1; the published introduction gives the displayed unified formula.
- M. V. Budrevich and A. E. Guterman, *Kräuter conjecture on permanents is true*, J. Combin. Theory A 162 (2019), 306–343 (paper; journal), for a different, resolved rank-dependent upper-bound conjecture.

Status check — 2026-09-10

Checked Ingram–Razborov's August 2026 published introduction, which explicitly retains the minimum-positive-value conjecture, and its preprint §6, Problem 3. Wanless (2005), §3, pp. 430–431, supplies attaining matrices for every $n \leq 20$, a proved parameter range of this target. Exact-title and minimum-positive-permanent searches found no general resolution. The 2019 Kräuter theorem proves a rank-dependent upper bound; Hunter–Kwan–Sauermann proves exponential range cardinality. Neither establishes general attainment of the lower bound.